Project: Worked examples
Title: Load budget, PSU, transformer and battery
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

The demand of a set of loads per supply (three-phase AC, single-phase AC, DC) and the minimum DC power supply, transformer and battery for it, each with a margin; the heat the loads dissipate for the enclosure calculation.
Demand per load: P_i·d_i·k_i; apparent power P_i·d_i·k_i/cosφ_i (arithmetic sum, conservative) Bus currents: I_3φ = S_3φ/(√3·U), I_1φ = S_1φ/U, I_DC = P_DC/U_DC; when a bus is chosen for the DC power supply its input P_DC/η_PSU (power factor 1) is added to that bus Heat dissipated inside the enclosure: Φ = Σ h_i·P_i·d_i·k_i, h_i = the share of the input power released inside (blank: the loss 1 − η_i; electronics 1; a machine outside the enclosure 0) PSU: P_PSU ≥ margin·P_DC/f_derate (f_derate from the manufacturer's curve at θ and altitude) Transformer: S_T ≥ margin·S for the running load; starting peak S_peak = S + max(m_i·P_i/cosφ_i − S_i) ≤ k_T·S_T (the starting load undiversified), so S_T ≥ S_peak/k_T for the start; the rating to buy is the larger of the two Battery: C = I_b·t/(DoD·η_b·k_temp·k_age), I_b = Σ backed P_i·d_i·k_i/(U_b·η_conv); Peukert: I_b·H·(t/H)^(1/k_p) in place of I_b·t
Ref: Linden's Handbook of Batteries 5th ed., Ch. 3–4 — Battery sizing and the Peukert relation
Ref: BS EN 60204-1:2018 cl. 4.3 — Electrical supply (supply voltage tolerance; cited by number only)

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| loads | Loads | 6 row(s) | | drawing: E-SCH-010 rev A | any |
| loads[1] | | name = Duty pump; supply = AC three-phase; P = 11 kW; cosphi = 0.86; eta = 0.91; d = 1; k = 1; inrush = 6.5; backed = no | | | |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| loads[2] | | name = Standby pump; supply = AC three-phase; P = 11 kW; cosphi = 0.86; eta = 0.91; d = 1; k = 0; inrush = 6.5; backed = no | | | |
| loads[3] | | name = Panel fan; supply = AC single-phase; P = 0.06 kW; cosphi = 0.9; eta = 0.7; d = 1; k = 1; inrush = 1; backed = no | | | |
| loads[4] | | name = Panel light; supply = AC single-phase; P = 0.02 kW; cosphi = 0.95; eta = 1; d = 0.1; k = 1; inrush = 1; backed = no | | | |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| loads[5] | | name = PLC and I/O; supply = DC; P = 0.096 kW; cosphi = 1; eta = 0.9; d = 1; k = 1; inrush = 1; backed = yes | | | |
| loads[6] | | name = Router; supply = DC; P = 0.024 kW; cosphi = 1; eta = 0.9; d = 1; k = 1; inrush = 1; backed = yes | | | |
| P_add | Additional three-phase load (linked from E1's RMS power) | 0 | kW | default | 0 … 200 kW |
| cosphi_add | Power factor of the additional load | 0.85 | | default | 0.5 … 1 |
| eta_add | Efficiency of the additional load | 0.9 | | default | 0.5 … 1 |
| inrush_add | Starting multiple of the additional load | 1 | | default | 1 … 20 |
| heat_add | Share of the additional load's input power released inside the enclosure (a linked machine is outside it: 0) | 0 | | default | 0 … 1 |
| U_3ph | Three-phase line voltage | 400 | V | spec: Site supply | 12 … 1000 V |
| U_1ph | Single-phase voltage | 230 | V | spec: Site supply | 12 … 1000 V |
| U_DC | DC bus voltage | 24 | V | drawing: E-SCH-010 rev A | 12 … 1000 V |
| psu_bus | AC bus that feeds the DC power supply (its input P_DC/η_PSU is added to that bus) | AC single-phase | | drawing: E-SCH-010 rev A | any |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| eta_psu | DC power supply efficiency (power factor taken as 1) | 0.9 | | assumption: 24 V supply efficiency 90 %, power factor taken as 1. | 0.5 … 1 |
| margin | Sizing margin | 1.25 | | assumption: 25 % growth margin. | 1 … 1.5 |
| theta | Ambient temperature | 45 | degC | spec: Plant room design maximum | -20 … 70 degC |
| f_derate | PSU derating factor at the ambient and altitude | 1 | | assumption: PSU rated to 60 °C without derating (datasheet). | 0.1 … 1 |
| P_PSU | Selected PSU rating | 240 | W | drawing: E-SCH-010 rev A | 1 … 100000 W |
| S_T | Selected transformer rating | 25 | kVA | spec: Existing site transformer allocation | 0.01 … 10000 kVA |
| k_T | Transformer short-time overload allowed for inrush | 3 | | assumption: Short-time overload allowed by the supply authority. | 1 … 20 |
| t_auto | Battery autonomy | 2 | h | spec: Telemetry to stay up for 2 h on mains loss | 0 … 72 h |
| U_b | Battery nominal voltage | 24 | V | drawing: E-SCH-010 rev A | 2 … 1000 V |
| eta_conv | Conversion efficiency battery → loads | 1 | | assumption: DC UPS module, loads on the battery bus. | 0.5 … 1 |
| DoD | Allowed depth of discharge | 0.8 | | assumption: VRLA, 80 % depth of discharge. | 0.1 … 1 |
| eta_b | Battery discharge efficiency | 0.95 | | assumption: Discharge efficiency. | 0.5 … 1 |
| k_temp | Temperature capacity factor | 1 | | assumption: Battery kept at 20 °C. | 0.3 … 1.2 |
| k_age | Ageing (end-of-life) factor | 0.8 | | assumption: End-of-life capacity 80 %. | 0.5 … 1 |
| C_batt | Selected battery capacity | 17 | A*h | drawing: E-SCH-010 rev A | 0.1 … 100000 A*h |

## 3. Assumptions

A1. Each load's P is its electrical input; DC loads count at power factor 1 in the transformer demand; with a

PSU bus chosen they count at P/η_PSU, the PSU's own loss being a load row if its heat is wanted. (Load budget convention.)
A2. The heat released inside the enclosure is h·P·d·k per load; the default h = 1 − η is the loss of a drive or supply, electronics release all their input (h = 1) and a machine outside the panel releases none (h = 0). (Green G-B-04.)
A3. Apparent powers are added arithmetically, which is conservative when power factors differ. (Upper bound on S.)
A4. PSU derating, transformer overload and battery factors are the manufacturer's values, entered as records. (§A3 reference records.)
A5. eta_psu (DC power supply efficiency (power factor taken as 1)) is taken as 0.9. (24 V supply efficiency 90 %, power factor taken as 1.)
A6. margin (Sizing margin) is taken as 1.25. (25 % growth margin.)
A7. f_derate (PSU derating factor at the ambient and altitude) is taken as 1. (PSU rated to 60 °C without derating (datasheet).)
A8. k_T (Transformer short-time overload allowed for inrush) is taken as 3. (Short-time overload allowed by the supply authority.)
A9. eta_conv (Conversion efficiency battery → loads) is taken as 1. (DC UPS module, loads on the battery bus.)
A10. DoD (Allowed depth of discharge) is taken as 0.8. (VRLA, 80 % depth of discharge.)
A11. eta_b (Battery discharge efficiency) is taken as 0.95. (Discharge efficiency.)
A12. k_temp (Temperature capacity factor) is taken as 1. (Battery kept at 20 °C.)
A13. k_age (Ageing (end-of-life) factor) is taken as 0.8. (End-of-life capacity 80 %.)

## 4. Working

[22] L1 = P·d·k; P·d·k/cos□
= 11 × 1 × 1 = 11 kW; 12.791 kVA
= 11 kW
Duty pump (AC three-phase)

[23] L2 = P·d·k; P·d·k/cos□
= 11 × 1 × 0 = 0 kW; 0 kVA
= 0 kW
Standby pump (AC three-phase)

[24] L3 = P·d·k; P·d·k/cos□
= 0.06 × 1 × 1 = 0.06 kW; 0.066667 kVA
= 0.06 kW
Panel fan (AC single-phase)

[25] L4 = P·d·k; P·d·k/cos□
= 0.02 × 0.1 × 1 = 0.002 kW; 0.0021053 kVA
= 0.002 kW
Panel light (AC single-phase)

[26] L5 = P·d·k; P·d·k/cos□
= 0.096 × 1 × 1 = 0.096 kW; 0.10667 kVA
= 0.096 kW
PLC and I/O (DC)

[27] L6 = P·d·k; P·d·k/cos□
= 0.024 × 1 × 1 = 0.024 kW; 0.026667 kVA
= 0.024 kW
Router (DC)

[28] P_conn = □ P_i
= 11 + 11 + 0.06 + 0.02 + 0.096 + 0.024
= 22.2 kW
Connected load

[29] P_dem = Σ P_i·d_i·k_i
= 11 + 0 + 0.06 + 0.002 + 0.096 + 0.024
= 11.182 kW
Demand active power
[30] S_dem = Σ P_i·d_i·k_i/cosφ_i
= 12.791 + 0 + 0.066667 + 0.0021053 + 0.10667 + 0.026667
= 12.993 kVA
Demand apparent power (arithmetic)
[31] I_3ph = S_3φ/(√3·U_3ph)
= 12791 / (√3 × 400)
= 18.462 A
Three-phase bus current
[32] I_1ph = S_1φ/U_1ph
= 202.11 / 230
= 0.87872 A
Single-phase bus current
[33] I_DC = P_DC/U_DC
= 120 / 24
= 5 A
DC bus current
[34] Phi = Σ h_i·P_i·d_i·k_i
= 0.09×11 + 0.09×0 + 0.3×0.06 + 0×0.002 + 0.1×0.096 + 0.1×0.024
= 1.02 kW
Heat dissipated inside the enclosure (→ E2)
[35] P_DC = Σ DC P_i·d_i·k_i
= 120 W
= 120 W
DC demand
[36] P_PSU_req = margin * P_DC / f_derate
= 1.25 × 120 W / 1
= 150 W
Minimum PSU rating
[37] S_T_req = margin * S_dem
= 1.25 × 12.993 kVA
= 16.241 kVA
Minimum transformer rating for the running load
[38] S_peak = S_dem + max(m_i·P_i/cosφ_i - S_i)
= 12.993 + 70.349 (Duty pump starting)
= 83.342 kVA
Peak apparent power while the largest load starts
[39] S_T_req_start = S_peak / k_T
= 83.342 kVA / 3
= 27.781 kVA
Minimum transformer rating for the starting peak
[40] S_T_min = max(S_T_req, S_T_req_start)
= max(16.241 kVA, 27.781 kVA)
= 27.781 kVA
Minimum transformer rating (running load and starting peak)
[41] I_b = Σ backed P_i·d_i·k_i/(U_b·η_conv)
= 120 / (24 × 1)
= 5 A
Battery discharge current

[42] Ah = I_b * t_auto
= 5 A × 2 h
= 10 A·h
Discharge capacity
[43] C_req = Ah / (DoD * eta_b * k_temp * k_age)
= 10 A·h / (0.8 × 0.95 × 1 × 0.8)
= 16.447 A·h
Minimum battery capacity
[45] u_PSU = P_PSU_req / P_PSU
= 150 W / 240 W
= 0.625
PSU utilisation
[47] u_T = S_T_req / S_T
= 16.241 kVA / 25 kVA
= 0.64964
Transformer utilisation
[48] u_inrush = S_peak / (k_T * S_T)
= 83.342 kVA / (3 × 25 kVA)
= 1.1112
Inrush utilisation
[50] u_batt = C_req / C_batt
= 16.447 A·h / 17 A·h
= 0.96749
Battery utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| P_conn | 22.2 kW | Connected load |
| P_dem | 11.182 kW | Demand active power |
| S_dem | 12.993 kVA | Demand apparent power |
| I_3ph | 18.462 A | Three-phase bus current |
| I_1ph | 0.87872 A | Single-phase bus current |
| I_DC | 5 A | DC bus current |
| Phi | 1.02 kW | Heat dissipated inside the enclosure |
| P_PSU_req | 150 W | Minimum PSU rating |
| S_T_req | 16.241 kVA | Minimum transformer rating, running load |
| S_peak | 83.342 kVA | Peak apparent power at start |
| S_T_req_start | 27.781 kVA | Minimum transformer rating, starting peak |
| S_T_min | 27.781 kVA | Minimum transformer rating, running and starting |
| I_b | 5 A | Battery discharge current |
| Ah | 10 A·h | Discharge capacity |
| C_req | 16.447 A·h | Minimum battery capacity |
| u_PSU | 0.625 | PSU utilisation |
| u_T | 0.64964 | Transformer utilisation |
| u_inrush | 1.1112 | Inrush utilisation |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| u_batt | 0.96749 | Battery utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Starting peak against transformer overload | u_inrush = 1.1112 | u_inrush ≤ 1 | 1.11 | |
| PASS | Battery capacity against required | u_batt = 0.96749 | u_batt ≤ 1 | 0.967 | |
| PASS | Transformer rating against required | u_T = 0.64964 | u_T ≤ 1 | 0.65 | |
| PASS | PSU rating against required | u_PSU = 0.625 | u_PSU ≤ 1 | 0.625 | |

## 7. Applicability

WARNING — record: Tabulated values not taken from a reference record: f_derate (PSU derating); DoD, eta_b, k_temp, k_age (Battery factors). Enter each as one so its source, edition and licence travel with it.
WARNING — input.P_add: P_add (Additional three-phase load (linked from E1's RMS power)) was not entered; the template default 0 kW is used.
WARNING — input.cosphi_add: cosphi_add (Power factor of the additional load) was not entered; the template default 0.85 is used.
WARNING — input.eta_add: eta_add (Efficiency of the additional load) was not entered; the template default 0.9 is used.
WARNING — input.inrush_add: inrush_add (Starting multiple of the additional load) was not entered; the template default 1 is used.
WARNING — input.heat_add: heat_add (Share of the additional load's input power released inside the enclosure (a linked machine is outside it: 0)) was not entered; the template default 0 is used.
WARNING — derate: f_derate: the ambient is above 40 °C but no PSU derating is entered; read the factor from the manufacturer's curve.

## 8. Verification

Template elec.load-budget@1.2.0 Content hash 82b6c9bbcbb22a37 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0